

```
/* =====  
DATA CONTROL LANGUAGE (DCL) LAB  
Microsoft SQL Server (MSSQL)  
Beginner to Intermediate Level  
eCommerce Security Use Case  
===== */
```

```
-- =====  
-- STEP 1: CREATE DATABASE  
-- =====
```

```
CREATE DATABASE ecommerce_security_db;  
GO
```

```
USE ecommerce_security_db;  
GO
```

```
/* =====  
STEP 2: CREATE SAMPLE TABLES  
===== */
```

```
CREATE TABLE products (  
    product_id INT IDENTITY(1,1) PRIMARY KEY,  
    product_name VARCHAR(100),  
    price DECIMAL(10,2),  
    stock_qty INT  
);  
GO
```

```
CREATE TABLE orders (  
    order_id INT IDENTITY(1,1) PRIMARY KEY,  
    customer_name VARCHAR(100),  
    total_amount DECIMAL(10,2)  
);  
GO
```

```
/* =====  
STEP 3: INSERT SAMPLE DATA  
===== */
```

```
INSERT INTO products  
(product_name, price, stock_qty)  
VALUES  
('Laptop', 1200000, 10),  
('Mouse', 25000, 100),  
('Keyboard', 45000, 50),  
('Monitor', 350000, 20),  
('Printer', 450000, 5);  
GO
```

```
INSERT INTO orders  
(customer_name, total_amount)  
VALUES  
('Aung Aung', 1250000),  
('Su Su', 45000),  
('Kyaw Kyaw', 300000);  
GO
```

/* =====

STEP 4: VIEW SAMPLE DATA

===== */

```
72  SELECT * FROM products;
73  GO
```

100 % No issues found

Results Messages

	product_id	product_name	price	stock_qty
1	1	Laptop	1200000.00	10
2	2	Mouse	25000.00	100
3	3	Keyboard	45000.00	50
4	4	Monitor	350000.00	20
5	5	Printer	450000.00	5

```
75  SELECT * FROM orders;
76  GO
```

100 % No issues found

Results Messages

	order_id	customer_name	total_amount
1	1	Aung Aung	1250000.00
2	2	Su Su	45000.00
3	3	Kyaw Kyaw	300000.00

/* =====

STEP 5: CREATE SQL SERVER LOGINS

NOTE:

Logins are created at SQL Server level

===== */

```
CREATE LOGIN admin_user
WITH PASSWORD = 'Admin@123';
GO
```

```
CREATE LOGIN sales_user
WITH PASSWORD = 'Sales@123';
GO
```

```
CREATE LOGIN report_user
WITH PASSWORD = 'Report@123';
GO
```

```
/* =====  
STEP 6: CREATE DATABASE USERS  
NOTE:  
Users are mapped to logins  
===== */
```

```
USE ecommerce_security_db;  
GO
```

```
CREATE USER admin_user  
FOR LOGIN admin_user;  
GO
```

```
CREATE USER sales_user  
FOR LOGIN sales_user;  
GO
```

```
CREATE USER report_user  
FOR LOGIN report_user;  
GO
```

```
-- check user roles
```

```
119 SELECT  
120     dp1.name AS database_role,  
121     dp2.name AS database_user  
122 FROM sys.database_role_members drm  
123 JOIN sys.database_principals dp1  
124     ON drm.role_principal_id = dp1.principal_id  
125 JOIN sys.database_principals dp2  
126     ON drm.member_principal_id = dp2.principal_id  
127 WHERE dp2.name = 'report_user';  
128
```

100 % No issues found

Results Messages

database_role	database_user
---------------	---------------

```

/* =====
STEP 7: VIEW USERS
===== */

```

```

133 SELECT
134     name,
135     type_desc
136 FROM sys.database_principals
137 WHERE type IN ('S', 'U');
138 GO
139

```

100 % No issues found

Results Messages

	name	type_desc
1	dbo	WINDOWS_USER
2	guest	SQL_USER
3	INFORMATION_SCHEMA	SQL_USER
4	sys	SQL_USER
5	admin_user	SQL_USER
6	sales_user	SQL_USER
7	report_user	SQL_USER

```

/* =====
STEP 8: GRANT PERMISSIONS
===== */

```

```

-- =====
-- ADMIN USER
-- Full control on database
-- =====

```

```

GRANT CONTROL
ON DATABASE::ecommerce_security_db
TO admin_user;
GO

```

```

/* =====
SALES USER
Can:
- Read products
- Read orders
- Insert orders
===== */

```

```
GRANT SELECT
ON products
TO sales_user;
GO
```

```
GRANT SELECT, INSERT
ON orders
TO sales_user;
GO
```

```
/* =====
REPORT USER
Read-only access
===== */
```

```
GRANT SELECT
ON products
TO report_user;
GO
```

```
GRANT SELECT
ON orders
TO report_user;
GO
```

```
/* =====  
STEP 9: TEST admin_user  
===== */
```

```
EXECUTE AS USER = 'admin_user';
```

```
GO
```

```
194 SELECT * FROM products;  
195 GO  
196
```

100 %  No issues found

	product_id	product_name	price	stock_qty
1	1	Laptop	1200000.00	10
2	2	Mouse	25000.00	100
3	3	Keyboard	45000.00	50
4	4	Monitor	350000.00	20
5	5	Printer	450000.00	5

```
197 SELECT * FROM orders;  
198 GO  
199
```

100 %  No issues found

	order_id	customer_name	total_amount
1	1	Aung Aung	1250000.00
2	2	Su Su	45000.00
3	3	Kyaw Kyaw	300000.00

```
REVERT;
```

```
GO
```

```
/* =====  
STEP 10: TEST sales_user  
===== */
```

```
EXECUTE AS USER = 'sales_user';  
GO
```

-- Allowed: View products

```
212 SELECT * FROM products;  
213 GO  
214
```

100 % No issues found

	product_id	product_name	price	stock_qty
1	1	Laptop	1200000.00	10
2	2	Mouse	25000.00	100
3	3	Keyboard	45000.00	50
4	4	Monitor	350000.00	20
5	5	Printer	450000.00	5

-- Allowed: View orders

```
217 SELECT * FROM orders;  
218 GO  
219
```

100 % No issues found

	order_id	customer_name	total_amount
1	1	Aung Aung	1250000.00
2	2	Su Su	45000.00
3	3	Kyaw Kyaw	300000.00

-- Allowed: Insert orders

```
INSERT INTO orders  
(customer_name, total_amount)  
VALUES  
('Mya Mya', 550000);  
GO
```



```

228     SELECT * FROM orders;
229     GO
230

```

100 % No issues found

	order_id	customer_name	total_amount
1	1	Aung Aung	1250000.00
2	2	Su Su	45000.00
3	3	Kyaw Kyaw	300000.00
4	4	Mya Mya	550000.00

REVERT;

GO

/* =====

STEP 11: TEST report_user

===== */

EXECUTE AS USER = 'report_user';

GO

-- Allowed: Read-only access

```

243     SELECT * FROM products;
244     GO
245

```

100 % No issues found

	product_id	product_name	price	stock_qty
1	1	Laptop	1200000.00	10
2	2	Mouse	25000.00	100
3	3	Keyboard	45000.00	50
4	4	Monitor	350000.00	20
5	5	Printer	450000.00	5

```

246     SELECT * FROM orders;
247     GO
248

```

100 % No issues found

	order_id	customer_name	total_amount
1	1	Aung Aung	1250000.00
2	2	Su Su	45000.00
3	3	Kyaw Kyaw	300000.00
4	4	Mya Mya	550000.00

REVERT;

GO

252	SELECT USER_NAME();
100 %	✓ No issues found
Results	Messages
	(No column name)
1	dbo

```
/* =====
STEP 12: TEST report_user CANNOT INSERT
===== */
```

```
EXECUTE AS USER = 'report_user';
GO
```

```
-- This should FAIL
```

262	INSERT INTO orders
263	(customer_name, total_amount)
264	VALUES
265	('Unauthorized User', 1000);
266	GO
100 %	✓ No issues found
Messages	
Msg 229, Level 14, State 5, Line 262	
The INSERT permission was denied on the object 'orders', database 'ecommerce_security_db', schema 'dbo'.	

```
-- REVERT back to dbo/admin
```

```
REVERT;
GO
```

```
/* =====
EXPECTED ERROR:
INSERT permission denied
===== */
```

```
/* =====  
STEP 13: REVOKE PERMISSION  
Remove INSERT permission from sales_user  
===== */
```

```
REVOKE INSERT  
ON orders  
FROM sales_user;  
GO
```

```
/* =====  
STEP 14: TEST REVOKE  
===== */
```

```
EXECUTE AS USER = 'sales_user';  
GO
```

-- This should FAIL after REVOKE

```
296  INSERT INTO orders  
297  (customer_name, total_amount)  
298  VALUES  
299  ('Blocked User', 5000);  
300  GO
```

100 %  No issues found

 Messages

Msg 229, Level 14, State 5, Line 296
The INSERT permission was denied on the object 'orders', database 'ecommerce_security_db', schema 'dbo'.

```
REVERT;  
GO
```

```
/* =====  
STEP 15: DENY DELETE PERMISSION  
===== */
```

```
DENY DELETE  
ON orders  
TO sales_user;  
GO
```

```
/* =====  
STEP 16: TEST DENY  
===== */
```

```
EXECUTE AS USER = 'sales_user';  
GO
```

-- This should FAIL

```
323  DELETE FROM orders  
324  WHERE order_id = 1;  
325  GO  
---
```

100 %   No issues found

 Messages

Msg 229, Level 14, State 5, Line 323
The DELETE permission was denied on the object 'orders', database 'ecommerce_security_db', schema 'dbo'.

```
REVERT;  
GO
```

```

/* =====
STEP 17: VIEW USER PERMISSIONS
===== */

```

```

334  SELECT
335
336      pr.name AS user_name,
337
338      pe.permission_name,
339
340      pe.state_desc,
341
342      ob.name AS object_name
343
344  FROM sys.database_permissions pe
345
346  JOIN sys.database_principals pr
347      ON pe.grantee_principal_id = pr.principal_id
348
349  JOIN sys.objects ob
350      ON pe.major_id = ob.object_id
351
352  ORDER BY pr.name;
353  GO

```

100 % No issues found

Results Messages

	user_name	permission_name	state_desc	object_name
1	report_user	SELECT	GRANT	products
2	report_user	SELECT	GRANT	orders
3	sales_user	SELECT	GRANT	products
4	sales_user	DELETE	DENY	orders
5	sales_user	SELECT	GRANT	orders

```

/* =====
STEP 18: VIEW CURRENT SESSION USER
===== */

```

```

359  SELECT
360      USER_NAME() AS current_database_user;
361  GO

```

100 % No issues found

Results Messages

	current_database_user
1	dbo

/* =====

STEP 19: VIEW CURRENT LOGIN

===== */

```
367  SELECT
368      SYSTEM_USER AS current_login;
369  GO
```

100 % No issues found

Results Messages

	current_login
1	May\May Yu Ya

/* =====

STEP 20: SHOW LOGIN INFORMATION

===== */

```
375  SELECT
376      name,
377      type_desc,
378      create_date
379  FROM sys.server_principals
380  WHERE type IN ('S', 'U');
381  GO
```

100 % No issues found

Results Messages

	name	type_desc	create_date
1	sa	SQL_LOGIN	2003-04-08 09:10:35.460
2	##MS_PolicyTsqlExecutionLogin##	SQL_LOGIN	2025-10-21 12:53:19.013
3	May\May Yu Ya	WINDOWS_LOGIN	2026-05-10 19:41:18.897
4	NT SERVICE\SQLWriter	WINDOWS_LOGIN	2026-05-10 19:41:18.903
5	NT SERVICE\Winmgmt	WINDOWS_LOGIN	2026-05-10 19:41:18.910
6	NT Service\MSSQL\$SQLEXPRESS	WINDOWS_LOGIN	2026-05-10 19:41:18.913
7	NT AUTHORITY\SYSTEM	WINDOWS_LOGIN	2026-05-10 19:41:18.923
8	NT SERVICE\SQLTELEMETRY\$SQLEXPRESS	WINDOWS_LOGIN	2026-05-10 19:41:20.270
9	##MS_PolicyEventProcessingLogin##	SQL_LOGIN	2026-05-16 13:55:26.213
10	admin_user	SQL_LOGIN	2026-05-21 10:04:12.563
11	sales_user	SQL_LOGIN	2026-05-21 10:04:15.410
12	report_user	SQL_LOGIN	2026-05-21 10:04:17.990

```
/* =====  
STEP 21: CLEANUP SECTION  
DROP USERS  
===== */
```

```
DROP USER admin_user;  
GO
```

```
DROP USER sales_user;  
GO
```

```
DROP USER report_user;  
GO
```

```
/* =====  
STEP 22: DROP LOGINS  
===== */
```

```
DROP LOGIN admin_user;  
GO
```

```
DROP LOGIN sales_user;  
GO
```

```
DROP LOGIN report_user;  
GO
```

```
/* =====  
END OF DCL LAB SCRIPT  
===== */
```